

Aktilek Skakova

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EDUCATION

University of Pennsylvania

Philadelphia, PA

Master's Program in Electrical Engineering (GPA: 3.67)

Jan 2025 – Dec 2026

- Coursework: PCB Design and Manufacturing, Microprocessor Systems, IoTs Sensors and Systems, Smart Devices, Electronic Circuits, DSA, Statistics for DS, Robots for Rehabilitation, IoT Edge Computing

Nazarbayev University

Astana, Kazakhstan

Bachelor of Electrical and Computer Engineering, (GPA: 3.59)

Aug 2020 – June 2024

EXPERIENCE

Electrical Engineer

Feb 2024 – Aug 2025

Alstom Electro-locomotives Company

Astana, Kazakhstan

- Delivered a **70%** reduction in **high-voltage (HV) battery** failure incidents by redesigning locomotive battery reconditioning procedures, restoring 600/1,000 cells and avoiding \$60k in replacement costs
- Led localization of **low voltage (LV) electrical systems** and **HVAC** maintenance by establishing in-house repair protocols; utilized **5 Whys**, **Ishikawa**, and **8D** methodologies for root cause analysis
- Conducted electrical **performance simulations** and validation of HVAC fan systems under operating conditions, contributing to a **50%** reduction in HVAC-related repair events increasing operational reliability
- Identified a **contactor-voltage mismatch** as the root cause of recurring RIOM malfunctions by analyzing fleet-wide failure data, preventing continuous RIOM activation and avoiding \$40–50k in downtime costs

Electrical Engineer Intern

May 2023 – Aug 2023

- Evaluated alternatives for obsolete **low-voltage (LV) electrical components** and localization, assessing compatibility and integration impact, and documenting results in a cross-referenced database using **SAP ERP**

PROJECTS

Interactive Word Construction Device (Custom Embedded PCB)

Jan 2026

- Designed, fabricated, and validated a custom embedded **PCB in Altium Designer** including power regulation, MCU integration, and **SPI / I2C** communication interfaces
- Programmed C-based control logic using binary-encoded **GPIO** to represent 26 letters with minimal pins, and designed a **SolidWorks** enclosure supporting full system integration and operation

Glove Based Motion Control System

Dec 2025

- Developed a real-time embedded human-machine interface using a 16-bit **IMU** and ATmega328P MCU, achieving sub-10 ms latency through **hybrid C and Python** firmware and signal processing
- Designed and optimized a low-latency embedded signal-processing stack (motion tracking, gesture inference, USB transport) running in 3 ms per frame on an **ATmega328P**
- Integrated sensors and peripherals using digital **communication interfaces** including I2C (OLED display) and SPI, enabling real-time visualization of IMU data and system status

Fruit Ripeness Detection System

May 2025

- Built an **embedded agricultural sensing device** integrating RGB and near-infrared sensors with a microcontroller to classify fruit maturity and deliver real-time LCD and audio feedback, developing a fully functional prototype that won 1st Place at the **Penn Engineering Hackathon**

Research: Applications of Signal Processing lab, ASP-LAB

Jan 2022 – June 2024

- Developed an adaptive Active Noise Control (ANC) system using **signal decomposition** and **digital filtering** techniques to suppress high-amplitude acoustic signals in MRI environments
- Implemented and evaluated adaptive control algorithms (FxLMS, FxNLMS, FxVSSLMS) in **MATLAB / Simulink**, achieving up to **37.4 dB** noise attenuation (80%) analyzing system stability and real-time performance

SKILLS

Programming: C, C++, Python, MATLAB, Git, ZephyrRTOS, MPLABX IDE, CODESYS, Linux

Software Applications: Altium Designer, LTSpice, SolidWorks, Simulink, Docker, AutoCAD, Simplicity Studio

Technical skills:: SPI, I2C, UART, Wi-Fi protocols, Embedded Firmware development, PCB fabrication, Power Architecture Design, Hardware Design, Circuit Debugging, Oscilloscope, Logic Analyzer, Microcontrollers, PLC systems